Characterization and optimization of droplet sorting technique in microfluidics

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Background

Microfluidic droplets are a very promising tool for biomedical and chemical applications, as they offer the possibility to compartmentalize and isolate reactions. At the same time, they also account for some of the most arduous models of fluid mechanics.

The goal of my internship was to characterize and optimize **single-cell** encapsulation of microfluidic water droplets in an oil medium, the objectives being to be able to control the size and the speed of the droplets, as well as maximize the number of droplets containing a cell. The generation of the droplet and the sorting are done in the same PDMS chip.

The parameters were the oil and water (containing the cells) flow rate or the design and dimensions of the chip.

Cell culture

- Use of HEK (Human Embryonic Kidney) cells because of their reliable growth
- The **cells confluency** has a very important impact on the experiments:
  - If too low, there are not enough cells entering the chips
  - If too high, the cells form groups and stick together which is bad for single-cell encapsulation
  - The targetted confluency for the experiments is 70-80%
  - The cells need to be kept alive even after the encapsulation in the droplets

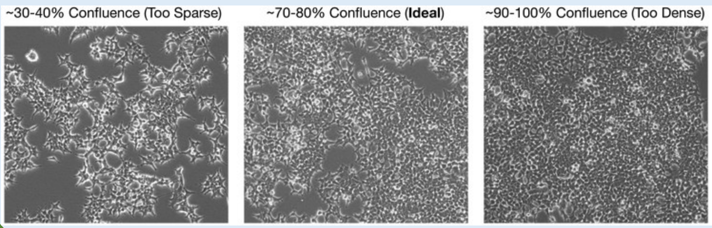

Figure 1: HEK cells confluency

Experimental Setup

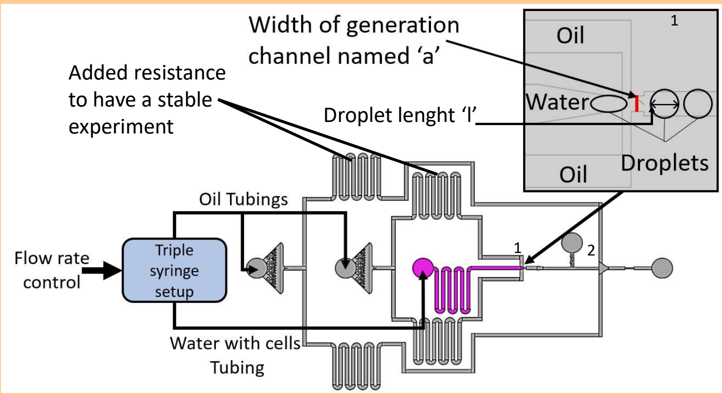

Figure 2: Experimental setup schematic

Why are there two inlets for the oil?

In Fig 2, there are two inlets for the oil , the second one is for the droplet generation while the second one has the purpose of spacing the droplets as well as controlling the speed of these droplets by injecting an oil flow in point 2 (Fig 2).

Main results

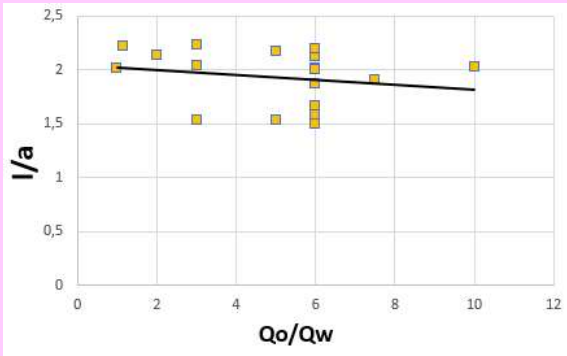

Figure 3: Dimensionless drop length l/a versus flow rate ratio Qw/Qo

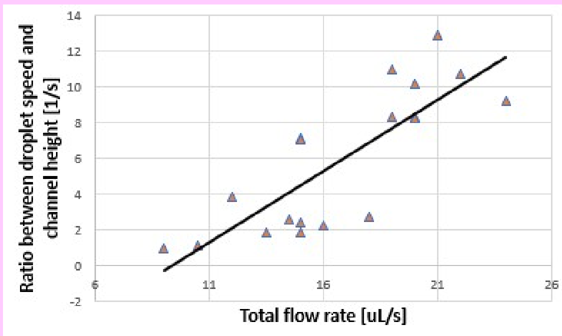

Figure 4: Droplet speed and channel height ratio versus total flow rate

- The ratio between the length and this width varies between 1,5 and 2,5 in figure 3, thus the control of the size of the droplet can be done by changing the width of the droplet generation channel. This gives an approximate size for the droplet, and the final size adjustment is done by slightly adjusting the water flow rate for the experiment.
- Different channel heights were used during the experiment, the channel surface hence the droplet speed is impacted which is why the ratio is used.
- The control of the oil flow rate in the first inlet (Figure 2) is the solution to control the speed of the droplet since it doesn't have an impact on the size of the droplets. The speed is proportional to the total flow rate.

Conclusion

Due to confidentiality matters, this poster unfortunately only shows a small part of what I did during my internship.

This internship was a very important and interesting experience and gave me a great overview of research in general. The hardest part was the preparation (taking care of the cells, manufacturing the chips) and planning of the experiments, which helped me greatly improve my organizational skills. The part that I preferred was the design of new chips to improve the experiments.